

Optimizing the Charging Station Placement By Considering the User's Charging Behavior

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Abstract—The successful introduction of electromobility relies considerably on the implementation of the charging stations. This implementation, should be cost effective and satisfy the energy demand of the electric vehicle users. This article presents a tool that computes the optimal charging infrastructure, by considering the placement and type of charging stations. To achieve this, we first calculate the spatiotemporal energy demand to account for the specific demand of each user. Next, we conduct a preselection step, where the locations and station types of little relevance are identified and excluded from optimization. The actual optimization step uses a multi-objective genetic algorithm with two objectives: minimizing the total installation costs of the infrastructure and minimizing the amount of trips that fail due to insufficient energy in the vehicles. Finally, the study analyzes two factors, which possibly influence the optimization algorithm: the user's charging behavior and developments of the battery energy efficiency.

Index Terms—Electric Vehicles, Charging Infrastructure, Cost Optimization, Genetic Algorithm.

I. INTRODUCTION

The placement of the charging infrastructure plays a vital role for the successful introduction of electric vehicles (EV). An optimal placed charging infrastructure will not only help to reduce psychological barriers to use electric vehicles, but also minimize the total costs for the introduction of electric mobility. Within the past decade, several studies explored different approaches to optimal dimension and placement of the required charging infrastructure when introducing electric vehicles.

Determining the charging energy demand plays an important role in this task. We calculate the energy demand by using the mobility model introduced in [1]. The model reflects the mobility and energy consumption of different modeled agents based on the Household Income and Travel Survey (HITS) of 2012. A similar statistics-based method to model the mobility behavior has been employed by Baouche [2], Andrews [3], and Chen [4]. The evaluated studies show different approaches to calculate the energy demand or a replacement for this. Chen [4] and Frade [5] calculate the energy demand using a

regression analysis. However, they assume a constant demand without considering the actual energy consumption of the agents. Chen, for example, only measures the parking time of the EVs without considering the actual energy consumption and demand of the vehicles. Cavadas [6] follows a similar approach using the car hours of parking slot occupation as the demand that needs to be maximized. This represent a constant energy consumption and demand for all vehicles. In contrast to these studies, we calculate the specific charging energy demand during every parking process for each of the respective agents.

After estimating the energy demand, we tackle the actual problem of locating the necessary charging stations (CS) to satisfy the demand. The optimal placement of charging stations is a typical maximum demand coverage problem, wherein the main objective is to cover a certain need by optimizing the placement of the facilities that satisfy the respective needs. These kind of problems might have more than one optimization goal. For example, maximizing the supply of the energy demand by the electric vehicles or minimizing the installation and maintenance costs of the charging stations. In the last decades, two main approaches have been widely used to solve these kind of problems: genetic algorithms (GA) and mixed integer programming (MIP). A selection of different approaches using these methods will be introduced here.

The MIP method has increasingly been used for the placement of charging stations within the last years. Some of the studies translate the maximum coverage problem into the minimization of the distance that every agent or user of the system has to travel every time they wish to charge their vehicles. This approach has been taken by Baouche [2] and Andrews [3]. Another approach is to maximize the supplied demand, whereby the demand is not necessarily the actual charging demand, as shown by Chen [4] and Cavadas [6].

The genetic algorithm is a meta-heuristic optimization procedure and is used to solve diverse real-world problems [7]. Several studies apply GAs in context of electric vehicle charging station placement problems. The majority focuses on the optimization of one sole objective by considering several constraints. The methodology of the evaluated studies differs mostly on the type of objective used to finally place the

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charging stations and the respective constraints. For instance, Dong [8] considers a maximum allowable budget per vehicle and generates a charging station network with diverse charging modes (CM). Furthermore, he rates the charging infrastructure according to the *failed trips*. Dong defines the *failed trips* – or missing trips – as those trips that agents cannot conduct or complete due to insufficient energy in their EVs. He finds the optimal location of the CS by minimizing the *failed trips* using the implemented GA. Other studies such as Li [9] and Jin [10] focus on the minimization of a cost oriented objective function, albeit with different goals. While Li assumes fixed energy demand points to determine the quantity and location of potential charging stations, Jin assumes a complete arbitrary energy demand and focuses on improving the genetic algorithm parameters to enhance the algorithm’s performance. Ge [11] introduces the CS placement based on the user’s minimum loss, which is defined as the weighted distance from certain arbitrary demand points to a potential charging station. Xi [12] analyzes the interactions between the optimization goal, maximize the number of vehicles that charge, and the constraint parameter, the available budget.

In context of this study, we optimize the location problem according to two objectives: the minimization of the total investment costs and the *failed trips*. In other words, we maximize the supply of energy demand for the different vehicles while minimizing the amount of charging stations. Since by nature these are two opposing optimization goals, we present a multi-dimensional Pareto-Front that will allow decision makers to visualize the scenarios for different budgets. The construction of the Pareto-Front follows the principles of the multi-objective genetic algorithm (MOGA), which Konak introduces in [13].

We conclude this article with a sensitivity analysis to identify the impact of the charging behavior, and of the battery characteristics on the charging station infrastructure.

II. METHODOLOGY

The presented charging station placement algorithm was conducted in three steps: The energy demand estimation, the location and charging mode (CM) preselection, and the actual optimization of the charging station distribution. Subsequently, we analyzed the influence of the charging behavior (CB) and the battery technology on the optimization process.

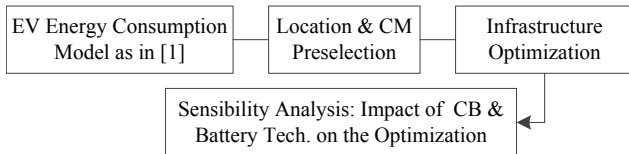


Fig. 1. Overview of the Optimization Process.

Here, we defined several variables and indices concerning the trip and optimization parameters. A summary of these parameters can be found in Table I.

TABLE I
LIST OF VARIABLES AND INDICES

Indices	Range	Description
a	$[1, n]$	Agents, EV users
CM	$\{m2, m3, m4\}$	Charging mode
d	$[1, k]$	Simulated days
l	$[0, L = 54]$	PA according to [14]
t	$[t_1, t_T]$	Time steps of simulation run
Variables	Description	
B	Number of charging stations per mode	
C	Charging station cost per mode	
C_{tot}	Total investment cost	
$E_{cons}(a, j)$	Energy consumption during trip j by agent a	
$E_{contBatt}(a, c)$	Nominal battery energy content of agent's a vehicle c	
$P_{cs,b}$	Charging power of the charging slot b	
$SOC(a, j)$	State Of Charge of agent a at the end of trip j	
η_{cs}	Efficiency of the charging station	
Δt_{park}	Parking duration	
$\omega_{fail}(a, d)$	Number of failed trips per agent and day	
Ω_{fail}	Total number of failed trips	

A. EV Energy Consumption Model

The first step contained the identification of a potential energy demand that needs to be satisfied by the optimized charging station infrastructure, which is proposed in II-C. Trippe and López Hidalgo [1] introduced an agent based mobility model that was used to calculate the energy demand. A detailed description of the model has been introduced in [1], such a description is outside of the scope of the present work as the present article constitutes an application of the energy consumption and demand model introduced by Trippe and López Hidalgo. Hence, this article presents only an overview of the energy consumption model.

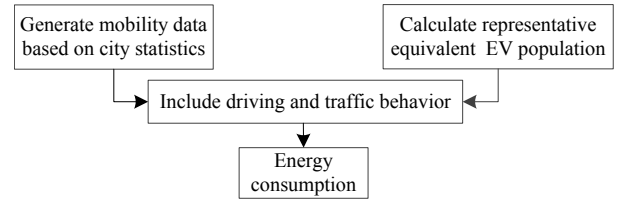


Fig. 2. Overview of the energy consumption and mobility model [1]

The mobility model presented in [1] computes the energy consumption of each EV in three main steps: In the first step, it generates mobility profiles using the Household Income and Travel Survey (HITS) conducted by the Land Transport Authority (LTA) of Singapore. The generated profiles are statistically representative and mirror the mobility behavior and driving habits of the Singapore’s drivers. In the second step, the model uses statistics on the current ICE-vehicle population of Singapore [15] and a list of currently commercially available electric vehicles to generate an analogue EV population. The resultant population considers different vehicle masses and nominal battery energy capacities. The nominal battery capacity is distributed among the vehicle

population to maintain an equivalent distribution to the current ICE-vehicle classes distribution by the LTA [15]. Table II shows the vehicles' characteristics and their allocation among the different classes.

In the third and final step, the agents' energy consumption will be calculated. The model considers different driving behavior patterns and traffic conditions throughout the day and calculates the energy consumption based on the power required to overcome the driving resistance. The resulting energy consumption is therefore specific to each agent and representative for the mobility and driving behavior in Singapore. This constitutes a considerable improvement in contrast to previous studies, which mainly considered a constant and homogeneous energy demand.

TABLE II
CHARACTERISTICS OF THE VEHICLE POPULATION FOR AN PENETRATION RATE OF 0.5% EV

Vehicle Class	1	2	3	4	5
Vehicle pop.	394	459	958	947	342
Avg. battery size in kWh	13	18	22	23.5	56
Median vehicle mass in kg	1117	1467	1500	1787	1943

In using the mobility model, we simulated the parking and driving activities of 3,100 vehicles – a 0.5% EV penetration rate – for a two-week period in Singapore. The results were 2.45 million driving and parking activities, which included the following information: start and stop location, start and stop time, trip energy consumption, and the parking purpose. Each location was allocated to one of 55 Planning Areas (PAs), a regional division introduced by the Singaporean Urban Redevelopment Authority. Each PA has a population of about 150,000 and is served by a town center and several neighborhood commercial/shopping centers [14]. These PAs represent potential locations for charging stations. The exact location of the charging slots within a PA will depend on several factors, specially on the availability of land where to place them and on the capacity of the low voltage grid. The generated activities can be divided into two types of activities: driving activities, which lead to an energy consumption, and parking activities, which provide the possibility to charge the vehicle. The spatiotemporal energy demand, which is vital for the placement of the charging stations, has been calculated during the parking activities. The charging behavior played a significant role in the calculation of the energy demand, since it represents the willingness of the agents to charge the vehicle and thus to refill the energy which has been consumed during previous trips. In the first step, we implemented a minor modification to the charging behavior model introduced in [1]. This behavior serves only as an illustration of a possible behavior and will be used later in this work as a reference charging behavior. The reference CB will help to identify the influence of the charging behavior of the different users on the optimization algorithm. The behavior model used the parking time, Δt_{park} , and the State of Charge (*SOC*) of the vehicle to determine whether the agents would charge their vehicles. The

charging decision was therefore completely independent of the availability of a charging station. The agents would attempt to charge by using following charging rules as introduced in [1]:

- parking time $\Delta t_{park} \geq 2 h$ and $SOC \leq 60\%$ or
- parking time $\Delta t_{park} \geq 5 min$ and $SOC \leq 20\%$

We assumed that the agents would not have to modify their behavior in order to charge the vehicle. Allowing the users to maintain their current driving and traveling behavior will certainly facilitate the introduction of electromobility. The agents would therefore adhere to their regular schedules, altering neither the time of their activities nor the parking locations. Therefore, charging at a station located at a different position was not allowed. Consequently, each vehicle could only decide to charge during the parking processes that had been predefined by using the mobility model of Trippe and López Hidalgo [1], wherein the authors demonstrated the calculation of the EV energy demand distribution. For this purpose, Trippe and López Hidalgo assumed that there was no limit to the number of charging stations available, that is, every attempted charging process was successful. In order to identify the optimal charging infrastructure, we linked the success of a charging process to the existence and availability of a charging station at the respective parking location. Section II-B introduces this process in detail.

Furthermore the proposed optimization considered three different charging modes: Mode 2, Mode 3, and Mode 4 as described in the IEC Standard 61851-1 [16]. The characteristics of the different modes are summarized in Table III. The CS investment costs include equipment, installation, and land acquisition costs. The prices are for a charging station with one charging slot. Due to the lack of space in Singapore, the majority of the costs arise from the land acquisition costs.

TABLE III
CHARGING MODE CHARACTERISTICS

Type	Mode	Power	Efficiency	Costs [17]
AC Slow	Mode 2: 1 ϕ	3 kW [18]	93% [18]	3,000 USD
AC Fast	Mode 3: 3 ϕ	22 kW [19]	88% [19]	30,000 USD
DC Fast	Mode 4	50 kW [19]	94% [19]	61,000 USD

B. Charging Station Location Selection

The charging slots placement assumed that PAs could include charging stations with different charging modes. This means that the optimization algorithm would have to examine 165 possible locations – three different charging modes (CM) and 55 planning areas. Hence, we introduced a location selection phase to reduce the computational load when optimizing the distribution of charging stations. The preselection identified important locations (PAs) and the charging modes to be used at the respective locations.

The main goal of the charging infrastructure is to enable vehicles to finish their trips successfully. Therefore, to achieve this goal we analyzed the contribution of each possible location, and thus determined the importance of each location.

The starting point is the limitless infrastructure approach introduced in [1]. Subsequently, the introduced 2.45 million activities are simulated 165 times. Each simulation run has been characterized by the placement of a limitless number of charging stations in 164 combinations while the remaining PA/CM combination, which was evaluated during the respective run, has been excluded. The rating of a location/mode combination was based on the number of trips, which the agents were not able to conclude successfully if the analyzed charging possibility would not be available. In order to determine the amount of unsuccessful trips, we updated for all the agents their individual *SOC*'s after each trip. When the vehicle's *SOC* dropped below 5% during a trip the respective trip was declared as a *failed trip*. The PA/CM combinations were then sorted and rated according to the total number of *failed trips* they caused when excluded. The more occasioned *failed trips*, the higher the rating and consequently the importance of the combination. Each simulation run followed the same schema: At the start of each parking process, the agents decided whether to charge or not by considering their charging behavior as introduced in Section II-A. For a vehicle to be able to charge, the vehicle's current position required at least one charging slot that was also available, i.e. it was not blocked by another driver. The agents selected the charging mode following the selection algorithm depicted in Figure 3.

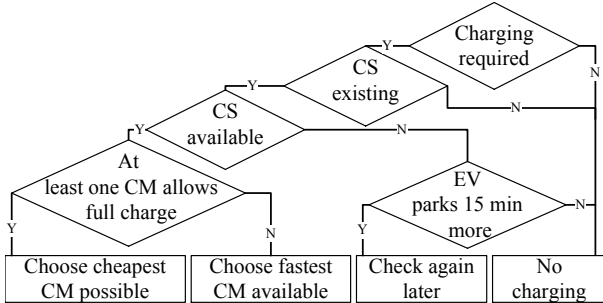


Fig. 3. Charging Station Allocation Process

The main priority of the agents was to reach a full battery during the charging process. When several charging modes were available, –one or more CM fulfilled $\Delta t_{park} P_{cs,b} \eta_{cs} \geq E_{contBatt}(a, c)$ –the agent selected the cheapest mode according to Table III. We assumed the charging costs to be proportional to the installation costs. If none of the available charging modes allowed a full charge, the driver selected the fastest mode available in order to maximize his *SOC*. At the end of the parking process, we updated the *SOC* of the agents according to the amount of energy they were able to charge. In the case there was no charging slot available –all slots were occupied by other users –the driver tried to charge fifteen minutes later as long as he had not reached the end of his parking time and the charging behavior rules still applied. If no charging slot was available during the parking time, the agents would try to charge at the next parking location and the selection process was repeated. This scenario, where the agents are able to attempt charging every fifteen minutes or at future

location is feasible by using any intelligent load management system as it has been introduced for instance by [20] and [21]. Failure to charge the vehicle at a specific charging location did not necessarily mean that the following trip would be a failed trip. Provided that the vehicle's *SOC* was enough to cover the trip and it was possible to charge at a future location, a failed charging process would not lead to a *failed trip*. Consequently, we focused on the *failed trips* and on the PA/CM combinations necessary to reduce the failed trips, rather than on scrutinizing the failed charging processes. A sole failed charging process is not necessarily crucial for the agents to be able to cover their mobility needs. This approach also allowed us to ignore the fact that some vehicles did not reach a full charge at certain locations. As long as the charging process helped to avoid a *failed trip*, the PA/CM combination would gain importance. At the same time, a combination that was highly frequent would not necessarily be an important combination. If the charging processes were not necessary for the agents to fulfill their mobility needs, the combination would lose importance.

The update of the agents' *SOC* at the end of the different trips, was conducted under consideration of the energy consumption during the last driving activity and the *SOC* of the respective vehicle, before starting the trip. We then calculated whether or not the trip had been successful –a *failed trip* occurred when the vehicle's *SOC* dropped below 5%. After each simulation run, the *failed trip* rate was then used to evaluate the importance of the charging station that was missing during the respective run. A PA/CM combination was considered to be low importance if the total number of *failed trips* did not increase when the PA/CM combination was excluded. Low importance combinations were excluded and were not considered in the optimization process in Section II-C. The important combinations were transferred to the CS placement process, which determines the number of stations in the relevant locations, and charging modes.

C. Charging Station Placement Optimization

The selection algorithm of II-B selected the most relevant locations for the placement of the charging stations. We then used a multi-objective genetic algorithm to optimize the distribution and amount of charging slots in the selected locations. The optimization goals included the minimization of the total investment costs of the charging stations (1) and the minimization of the *failed trips* for all users (4). The investment costs, C_{tot} were calculated as the sum of charging station costs for each mode C_{CM} multiplied by the number of placed CS B_{CM} in the respective charging modes.

$$\min C_{tot} = B_{m2} \cdot C_{m2} + B_{m3} \cdot C_{m3} + B_{m4} \cdot C_{m4} \quad (1)$$

The number of charging slots to be installed was limited by the number of slots necessary to satisfy all charging wishes, $B_{CM,max}$, i.e. the required amount when a limitless charging infrastructure was assumed. The boundaries of the amount of charging slots for each mode were defined as:

$$0 \leq B_{CM} \leq B_{CM,max} \quad (2)$$

The sum of the number of charging slots in each planning area determined the final amount of charging stations per mode:

$$B_{CM} = \sum_{l=0}^L B_{CM}(l) \quad (3)$$

The second optimization goal was the total number of *failed trips* Ω_{fail} . The *failed trips* measured the effectiveness of the charging station network. We defined the effectiveness as the sum of *failed trips* per agent and day $\omega_{fail}(a, d)$ during the observed period. The second minimization was stipulated as:

$$\min \quad \Omega_{fail} = \sum_{a=1}^n \sum_{d=1}^k \omega_{fail}(a, d) \quad (4)$$

The break-off criteria for the algorithm were defined by the number of generations and the population size. The population size defines how many different combinations of charging stations are evaluated per generation. The number of generations represents the evolutionary steps, i.e. iterations. Grefenstette [22] optimizes the GA's parameter settings based on a large set of test problems. Based on these parameter settings, we initially set the population size and the number of generations to 80 and 21 respectively. However, the results obtained with these settings did not converge. Therefore, we extended the number of generations to 300. The result of the optimization process was a Pareto-Front with a range of possible charging infrastructure solutions (Fig. 4). Each solution describes the optimal result - represented by a minimum of *failed trips* - for a certain investment budget.

D. Sensitivity Analysis

As introduced in Section II-A, the actual energy demand arises when the agents decide to charge their EVs. Only then, the energy consumption turns into an energy demand. Therefore, the charging behavior has a direct influence on the spatiotemporal energy demand and thus on the placement of the charging stations. The influence of the CB on the energy demand, has been addressed by Hahnel [20] and Franke [23]. Franke et. al. introduced two basic types of charging behavior. They demonstrated that electric vehicle users might be classified into two types based on the User Battery Interaction Style (UBIS): Some users will charge by following the behavior of their battery (e.g. when the SOC reaches a certain level –high UBIS), while other users will follow more their own behavior and charge for instance every time they are at a certain location or after a certain period of time –low UBIS. To analyze the influence of the CB on the optimization, we conducted the optimization steps using two different charging behaviors. The analyzed behaviors reflected different type of users as well as different willingness to take risks. The characteristics of the behaviors are summarized in Table IV.

Users of EVs with a high UBIS behavior are obviously influenced by the amount of energy available to them. The battery size can only be increased up to a certain physical

TABLE IV
CHARGING DECISION REQUIREMENTS FOR DIFFERENT SCENARIOS

Scenario	Parameter	Parking Activity Purpose			
		Home	Work & Others		
Risky	SOC	$\leq 90\%$	$\leq 10\%$	or	$\leq 50\%$
	Δt_{park}	$\geq 5min$	$\geq 5min$		$\geq 3h$
Prudent	SOC	$\leq 100\%$	$\leq 30\%$	or	$\leq 70\%$
	Δt_{park}	$\geq 5min$	$\geq 5min$		$\geq 1h$

limit. Therefore the energy density of the battery is the decisive factor. Matz [24] predicts an increase in the energy density of up to 200 Wh/kg on system level by 2020. We analyzed the influence of the battery energy density on the CS placement to account for improvements on the battery technology. We conducted the optimization step for 6 different energy density scenarios ranging from 120 to 250 Wh/kg by using the reference CB taken from section II-A. The battery capacities were not modified, and thus the vehicle weight and the specific energy consumption decreased with the increasing energy density.

III. RESULTS AND DISCUSSION

The charging station optimization routine was executed for a 0.5% EV penetration rate –3,100 EVs –by considering four different battery technologies. The agents charged their vehicles according to the charging behavior rules from section II-A. Furthermore, we analyzed the influence of the charging behavior and conducted the optimization by using the behavior rules introduced in Table IV, and by using batteries with an energy density of 200 Wh/kg . The resulting Pareto efficiency curves are plotted in Fig. 4.

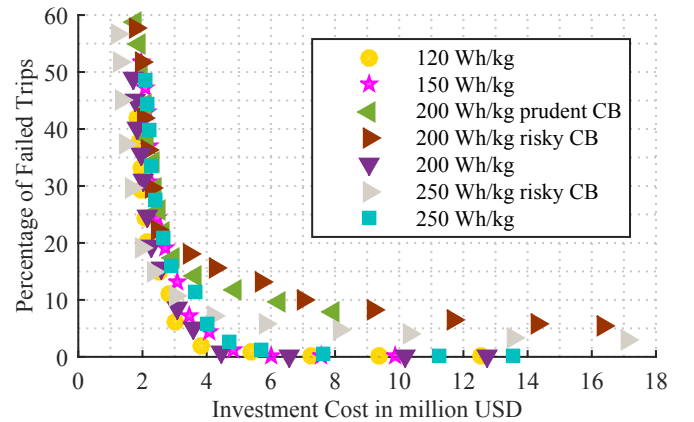


Fig. 4. Resulting Pareto-Fronts for the charging station optimization by considering different battery energy densities and charging behaviors. Results for an EV population of 3,100 vehicles.

The changes caused by a higher battery energy density are only minimal although the best results are reached with an energy density of 200 Wh/kg . The influence of the charging behavior is clearly noticeable. Specially, the risky CB has a noticeable number of failed trips independent of the charging

infrastructure. A probable reason for this is the low threshold for charging the battery -10% . Table V shows for comparisons the quantiles of the energy consumption per trip. For all battery technologies, the 75% percentile is between 2.6 and 2.7 kWh/trip. This is considerably higher than 10%*SOC* when considering batteries between 18 and 23.5 kWh, which are the majority of the vehicle population. Therefore, when an agent has two subsequent trips with a higher consumption, this will most probably lead to a failed trip.

TABLE V
QUANTILES OF THE ENERGY CONSUMPTION PER TRIP IN KWH FOR
DIFFERENT BATTERY ENERGY DENSITIES

	120 Wh/kg	150 Wh/kg	200 Wh/kg	250 Wh/kg
25 percentile	0.77	0.76	0.74	0.73
50 percentile	1.61	1.59	1.57	1.55
75 percentile	2.74	2.7	2.66	2.62

The development of the failed trip rate along with increasing infrastructure costs can be very well appreciated as shown in Fig. 4. The slope of the Pareto-Front is very moderate after a certain number of stations have been installed. Therefore, improving the *failed trips* ratio (FTR) is highly costly when trying to reach a FTR lower than 1%. We then analyzed closer the scenario with a battery energy density of 200Wh/kg. The best possible result in this case was a FTR of 0.08%, which has costs *USD* 18.35 m. We then evaluated all the combinations that lead to a FTR of 1% or lower and selected the most affordable combination. The selected combination allows for a FTR of 0.84% and costs *USD* 4.48 m. Table VI shows a comparison of the costs and number of charging stations for the combinations, which provide a FTR of 0.84% and 0.08%.

TABLE VI
CHARGING INFRASTRUCTURE COSTS IN USD AND REQUIRED AMOUNT OF
CHARGING SLOTS FOR DIFFERENT FAILED TRIP RATES (FTR)

	Total	Mode 2	Mode 3	Mode 4
FTR 0.84% Slots	310	212	69	29
FTR 0.84% Costs	4.48 m	0.64 m.	2.07 m.	1.77 m.
FTR 0.08% Slots	759	363	220	176
FTR 0.08% Costs	18.35 m.	1.09 m.	6.6 m.	10.74 m.

The exact distribution of the charging stations among the different PA for a FTR of 2.24% is plotted in Figure 5. The distribution clearly shows how some PA/CM combinations are neglected and as a result of the preselection step are left without station. Some PAs are left without any charging stations at all. This is for example the case for PA 8 –Western Water Catchment, and PA 16 –N.E. Islands, which are national parks are not usually visited by the same user on a regular basis. In this example, the planning area with the most stations is –Queenstown, which contains residential buildings, some industrial zones and several commercial buildings and shopping centers. The zones to the east of the country, Bedok

and Tampines registered also a very high demand. This areas are highly residential and in the case of Tampines, the parking durations are not very high, which would account for the amount of Mode 3 slots. It should be taken into account, that due to the heuristic nature of the GA the distribution of the charging stations among the PA is always different for each simulation run. The resulting Pareto-Fronts are always similar, so that reaching a local minimum can be excluded. However, the distribution of the charging stations among the PA/CM combinations might deliver very different results for each independent optimization routine. Therefore, the policymakers and city planers will have to evaluate and select an appropriate distribution by using their knowledge of the city.

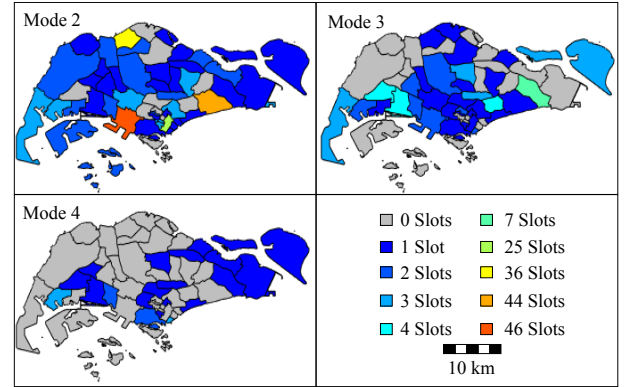


Fig. 5. Charging Slots Distribution for a failed trips rate of 0.84 % differentiated by charging modes. Results for an energy density of 200 Wh/kg and a EV population of 3,100 vehicles

Trippe and López Hidalgo [1] demonstrate that the results of the employed mobility model are scalable for different penetration rates of EVs. We examined the transferability of the results by conducting the optimization for 31,000 EVs –equivalent to an EV penetration rate of 5%. We consider the same four battery technologies, although the scrutiny of different charging behaviors was omitted in this step. The resulting Pareto-Fronts are depicted in Fig. 6. The resulting slope is marginally more moderate, than the slope when optimizing for 3100 EV. Furthermore, the stable FTR is slightly higher – 2.8%. Nonetheless, the results show a clear scalability trend for the optimization of different EV penetration rates, the variations arise from changes on the number of generations, which were necessary in order to compute the algorithm within a reasonable time.

IV. CONCLUSION AND OUTLOOK

In this article, we optimized the placement of charging stations for electric vehicles by considering a realistic energy demand for every user. The energy demand model considers the mobility behavior, a heterogeneous vehicle population, and different charging behaviors. Furthermore, we included a preselection step in order to identify the regions that were really relevant when optimizing the placement of charging stations. After conducting the optimization, we analyzed the influence of the charging behavior and the improvements of the

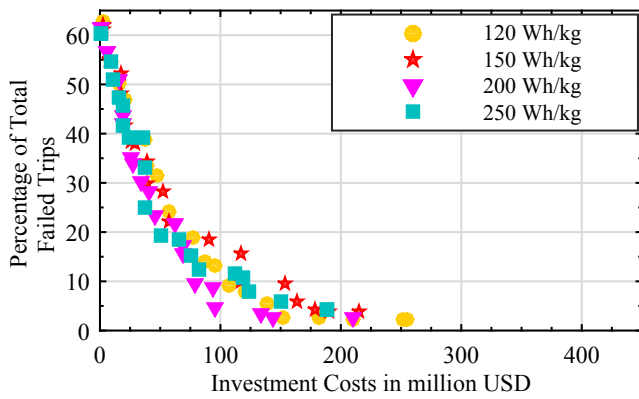


Fig. 6. Resulting Pareto-Fronts for the charging station optimization under consideration of different battery energy densities. Results for an EV population of 31000 vehicles

battery technology on the optimization. The results show that the charging behavior of the users is a major influencing factor when charging stations are placed. The resulting Pareto-Front selects the set of solutions, which minimizes the infrastructure investment costs, and the amount of failed trips. The analysis also considers predictions for different scenarios of the battery energy density and shows a scalability for different EV penetration rates. The results of this study can be used by policymakers to determine the required investment budget to satisfy the charging demand of a certain number of EV drivers and the characteristics of the required infrastructure: charging modes, locations and number of stations. Further research will be conducted to analyze in detail the different charging strategies, which are necessary for successfully introducing electric mobility. At present, there are unfortunately no data available on the usage of charging stations in Singapore. Thus, the obtained results still need to be validated.

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